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END SEMESTER EXAMINATION JUNE 2025

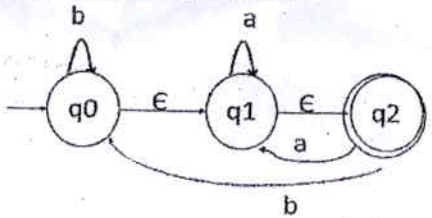
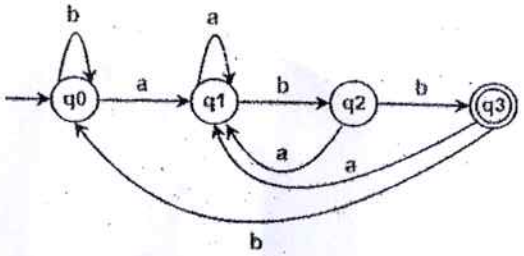
Name of the Program: **B. Tech** | Semester: **IV**

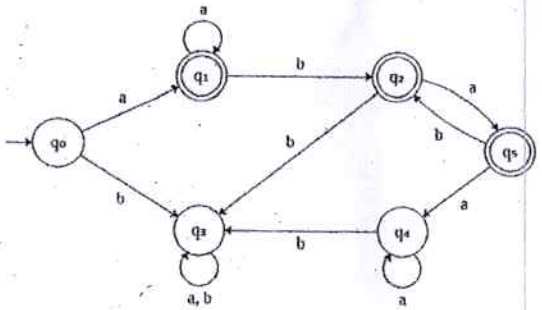
Name of the Course: **Finite Automata & Formal Languages** | Course Code: **TCS-402**

Time: 3 Hours | **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

Q1	**(20 Marks)**	
(a)	Design a DFA over the input alphabet $\sum = \{0, 1, 2\}$ that accept strings of 0's, 1's and 2's beginning with 0 followed by odd number of 1's and ending with even number of 2's.	CO2
(b)	Design a DFA over the alphabet $\sum = \{0, 1\}$ that accepts binary strings having a substring "00" **AND** "01".	CO2
(c)	Design a Moore machine for the rising edge detector system. This machine generates a short one-clock cycle pulse ***T*** when the signal level changes from 0 to 1, otherwise generates a ***D***. Convert the resultant Moore machine to Mealy machine and test the equivalence on the input "***101001***".	CO6
Q2	**(20 Marks)**	
(a)	Define ε -closure. Convert the given ε -NFA to DFA: 	CO2
(b)	State Arden's theorem. Find the regular expression for the language represented by the following DFA: 	CO2

(c)	Apply Myhill-Nerode theorem to minimize the given DFA: 	CO2
Q3	**(20 Marks)**	
(a)	State pumping lemma for context free languages (CFL). Prove that the given language 'L' is not a CFL: $L = \{ww \mid w \in (a, b)^*\}$	CO3
(b)	Construct a deterministic PDA that accepts all the strings from the given language using "acceptance by empty stack": $L = \{0^a 1^b 2^b 3^a \mid a>=1 \text{ and } b>=2\}$ Show that the resultant DPDA accepts the string "0001122333".	CO4
(c)	Convert the following CFG to Greibach normal form (GNF): $S \rightarrow CA \mid BB$ $A \rightarrow a$ $B \rightarrow b \mid SB$ $C \rightarrow c$	CO3
Q4	**(20 Marks)**	
(a)	Convert the following CFG to PDA: $S \rightarrow aA$ $A \rightarrow aABC \mid bB \mid a$ $B \rightarrow b$ $C \rightarrow c$ Also, test whether the string **"aaabc"** would be accepted/ rejected by the resultant PDA?	CO4
(b)	Define Ambiguity in CFG. Prove that the given CFG is ambiguous: $S \rightarrow SS \mid AB$ $A \rightarrow A0 \mid 0$ $B \rightarrow B1 \mid 1$	CO3
(c)	Simplify the given CFG by removing useless, null, and unit productions: $S \rightarrow ABCd \mid D \mid ab$ $A \rightarrow BC \mid \varepsilon$ $B \rightarrow bB \mid C \mid \varepsilon$	CO3

	$C \rightarrow cC \mid c$ $D \rightarrow dD$	
Q5	**(20 Marks)**	
(a)	Define Turing machine. Construct a Turing Machine for the given language over the input alphabet $\sum = \{0, 1, 2\}$: $L = \{0^m 1^n 2^{m+n} \mid m>=1 \text{ and } n>=1\}$	CO5
(b)	Design a Turing machine that works as a copier and copies the string of 0's and 1's. [Example: If the input string w is 01100 then the output will be ww that is 0110001100]	CO6
(c)	Explain the following (any two): (i) Post-correspondence problem (ii) Halting problem of Turing machine (iii) Universal Turing machine	CO6